

Machine Perception
Project 1
John Brooks

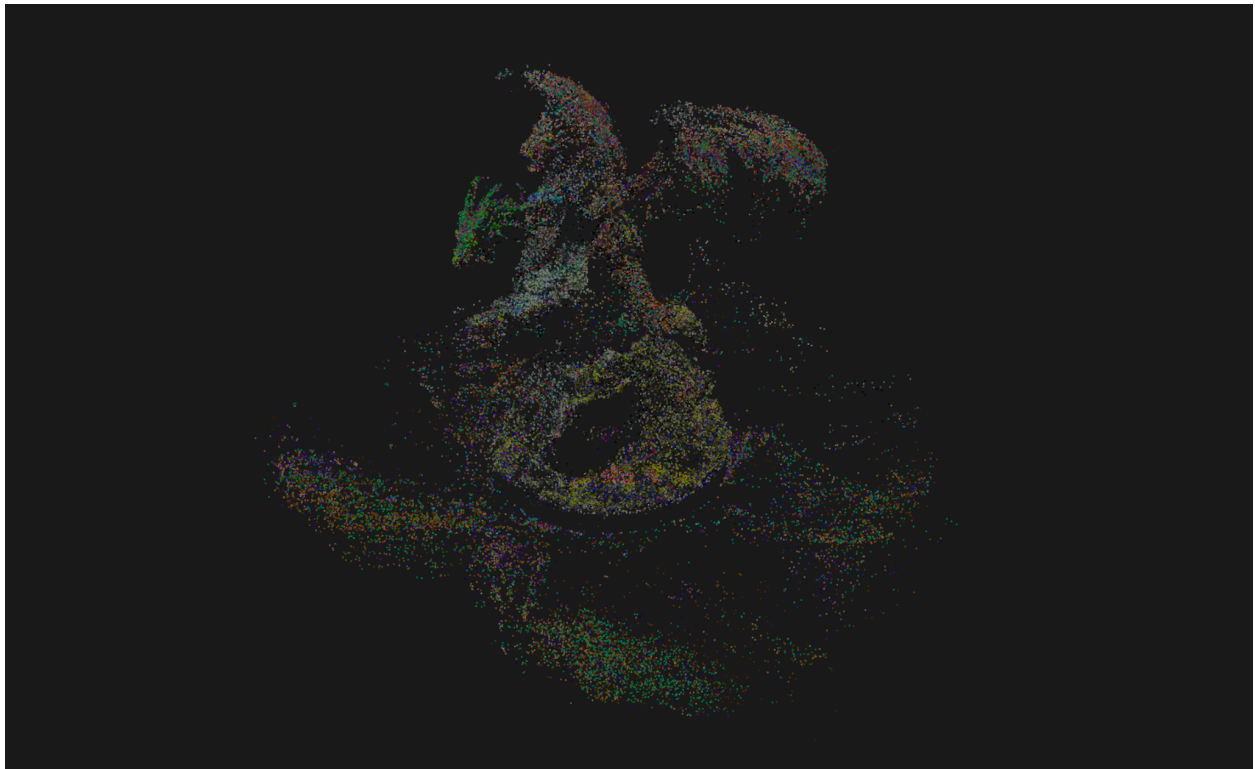
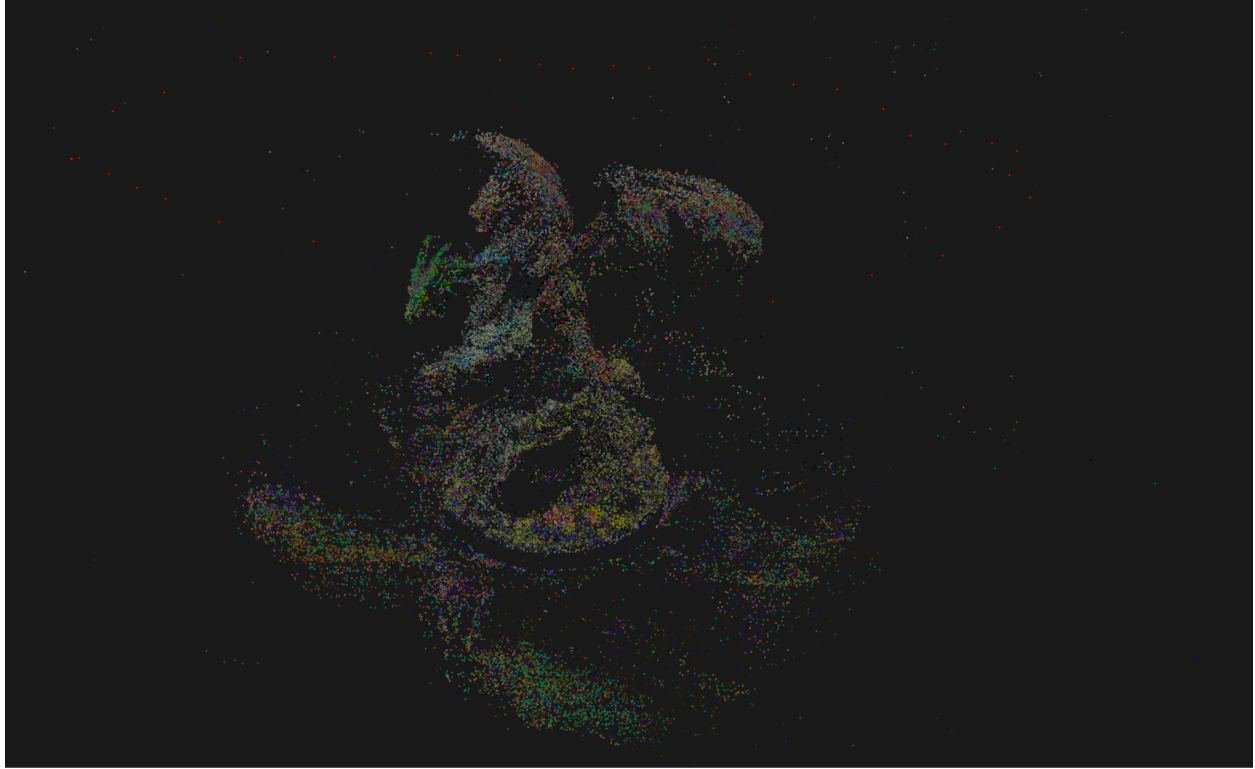
The i-sfm codebase that my project builds off of implements the incremental SfM algorithm. Firstly the dataset is loaded and SIFT features are extracted ([matching.py](#) line 44). Pairwise correspondences are formed with KNN matching and filtered by Lowe's ratio to keep only geometrically plausible pairs ([matching.py](#) lines 65-68). The matches are then pruned by a RANSAC fundamental matrix, producing an adjacency of image pairs that have enough inliers to be considered for reconstruction ([matching.py](#) lines 48-57). The system scores candidate baselines by estimating an essential matrix and pose for each connected pair and measuring parallax; the best pair seeds the construction. The chosen baseline is then initialized by recovering relative pose and triangulating inlier correspondences to produce the first batch of 3D points ([reconstruction.py](#) line 53-115).

From there, the pipeline grows the model incrementally with new 3D structure created by triangulating across views using the current camera pose, intrinsics, and depth checks ([reconstruction.py](#) lines 128-184). Global consistency is enforced by bundle adjustment that packs all camera pose parameters and point coordinates into a single vector, projects points with OpenCV, and minimizes reprojection error using a sparse Jacobian and robust loss ([bundle_adjustment.py](#) line 79). Utilities support calibration operations like loading the 3x3 K matrix for calibrating based on the camera ([utilis.py](#) lines 31-43), and results are rendered with Open3D by plotting camera frustrums and a colored point cloud ([visualizate_sfm.py](#) lines 10-28).

I collected three datasets for this project with only the final one producing reasonable results. The first dataset was of multiple stacked silicone lego bricks on a tile floor. While the SfM pipeline did complete it only captured feature matches of the floor and none of the brick itself. The second dataset was of a single brick on a white canvas to avoid background noise but during a run of the pipeline it failed due to low parallax. The final dataset of a painted 5 headed dragon sculpture successfully completed. The final working dataset contains 40 images taking in a circular pattern around the sculpture.

Overall the performance of the successful reconstruction was admirable as three out of the four heads are reasonably discernible and you can somewhat make out the fourth. The body of the dragon was not reconstructed well which might be attributed to the camera angle. Utilizing two python scripts [cleanup.py](#) and [measure_ply_object.py](#) I was able to measure the ratio of the Z axis to X or Y (whichever is larger ie. the wingspan) and got an output of 0.904 height/width ratio. From my ruler measurements the dragons actual height to wingspan ratio is 0.92 which is impressively close. Below you can see the dragon in the picture, the scanned dragon, and it cleaned up.





In summation, the pipeline goes as follows: feature extraction using SIFT, matching with Lowe's ratio test and RANSAC matrix estimation, camera pose estimation via PnP, and bundle adjustment for optimization. I think overall the pipeline needs considerable work as while the

success of the final dataset was impressive it should not have failed so miserably at the other two datasets. All the code and datasets for this project can be found on the following public git repo: <https://github.com/Johnmakuta/i-sfm>